

Assignment 1

Wings of Diversity

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Introduction

Butterflies are flying insects with scaled wings. Butterflies are important as they play a vital role in pollinating plants, ensuring the reproduction of fruits and vegetables, their population also shows the health of the ecosystem. In our report “Wings of diversity” the aim is to analyse which butterfly species were found in a specific site and on which plant species within Melbourne city. From this report we also able to find out that which site has the greatest number of butterflies.

Methods

Authors, Holly Kirk, Tessa Smith, Anna Backstrom, Alejandra Morán-Ordóñez, Georgia E. Garrard, Ascelin Gordon, Christopher D. Ives, Sarah Bekessy & Luis Mata. “Our City’s Little Gems – Butterfly diversity and flower-butterfly interactions in the City of Melbourne”, 2017

The dataset that we are using, has a total of 146012 observations. The observations were collected from various sites in Melbourne city for various butterfly species and their flower-butterfly interaction for several locations within the site. Some observations in the dataset aren’t that descriptive so that came as a limitation while analysing the data.

For this report we have used MS-EXCEL, using pivot-table, pivot-charts, and excel functions we have done the analysis.

Results

Answer 1:

QUESTION - 1									
site	Royal Park								
Sum of hum	LOCATION								
RELATIVE HUMI	1	2	3	4	5	6	7	8	9 Grand Total
0-10			0		16.4				16.4
10-20							613.6		223.6 837.2
20-30	533.2	1228	431.2	1539.2	926		744.4		1947.2 7349.2
30-40	120	778.8	4742	2965.6	982.4	688.4	1604.4	1719.2	2595.2 16196
40-50	891.6	720	3877.2	547.6	1390.8	2491.6	882.8	2087.2	1080 13968.8
50-60			838.8			1075.6			1914.4
Grand Total	1544.8	2726.8	9889.2	5052.4	3315.6	4255.6	3845.2	3806.4	5846 40282

Figure 1

Inference: The distribution of relative humidity levels at various sites within the Royal Park site was shown using the PivotTable analysis. The areas with the greatest total relative humidity readings were 3 and 7, whereas the places with moderate to low readings were 1, 2, 5, and 8.

This variation in relative humidity levels suggests that there may be variations in the microclimate and environmental conditions in Royal Park, which may have an impact on the suitability of the habitat for butterflies and the composition of the plants.

Answer 2:

QUESTION - 2					
Count of prap	WALK_NUM				
PLANT_FAMILY	1	2	3	4	Grand Total
Brassicaceae	52	32	16	12	112
Pittosporaceae	48	24	24	16	112
Campanulaceae	36	28	40	12	116
Goodeniaceae	55	40	17	12	124
Anacardiaceae	102	28	4	6	140
Plumbaginaceae	80	39	20	9	148
Myrtaceae	68	52	20	12	152
Lamiaceae	223	104	92	25	444
Fabaceae	294	143	76	31	544
Asteraceae	330	148	118	48	644
Grand Total	1288	638	427	183	2536

Figure 2

Inference: Pieris rapae butterfly are most found in Asteraceae plant family, even a high count of the same species of butterfly was found in Fabaceae plant family

as well! From our pivot table we can see that the greatest number of butterflies were spotted within the walk 1.

Answer 3:



Figure 3

Formula used: =SUMIFS(G:G,H:H,H11,X:X,X15)-SUMIFS(G:G,H:H,H15,X:X,X30,B:B,B10)

Answer 4:

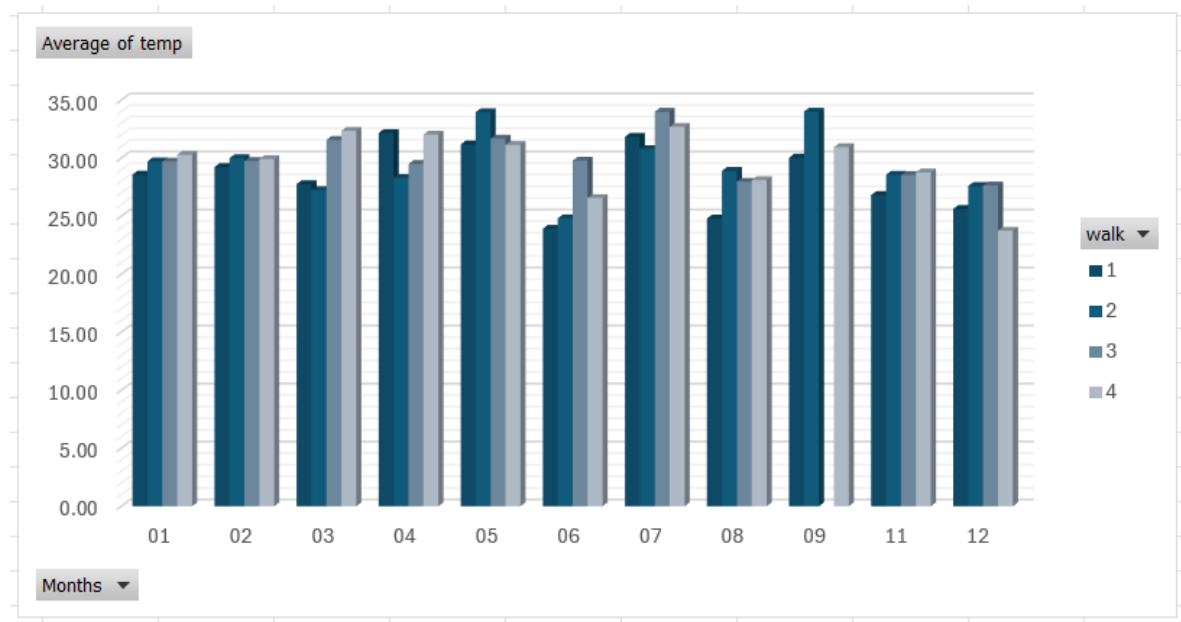


Figure 4

Inference: Month-to-month variations in the average temperature reveal seasonal trends in the study period's temperature. The highest average temperature is 32.29°C in July, with 32.15°C in May and 31.27°C in April. The lowest average temperature is recorded in December (23.75°C) with June (23.92°C), following closely behind. The average monthly temperature varies, suggesting the possibility of microclimatic variations or weather swings.

Answer 5:

SUM OF WALK IN A SITE PER LOCATION WITHIN THE SITE										
Sum of walk	LOCATION									
SITE_NAME		1	2	3	4	5	6	7	8	9 Grand Total
Argyle Square		415	151							566
Canning/Neill St Reserve		46								46
Carlton Gardens South		480	136	388						1004
Fitzroy-Treasury Gardens		164	252	168	205	77				866
Gardiner Reserve		44								44
Garrard Street Reserve		44								44
Lincoln Square		144	92							236
Murchinson Square		52								52
Pleasance Gardens		72								72
Princes Park		60	184	24	92	16	64			440
Royal Park		64	184	576	348	168	208	224	192	2360
State Library of Victoria		144								144
University Square		12	8							24
Westgate Park		216	241	140	280	273				1150
Womens Peace Gardens		124	136							260
Grand Total		2081	1384	1296	925	534	272	224	192	7308

Figure 5

Inference: There exists inter-site diversity in walk times, with certain sites registering longer walks for observation purposes than others. Longer walk lengths may be associated with larger ecosystems or a greater variety of flora types, which would offer a greater variety of opportunities to see butterflies. Sites with longer walk times overall might be hotspots for butterflies.

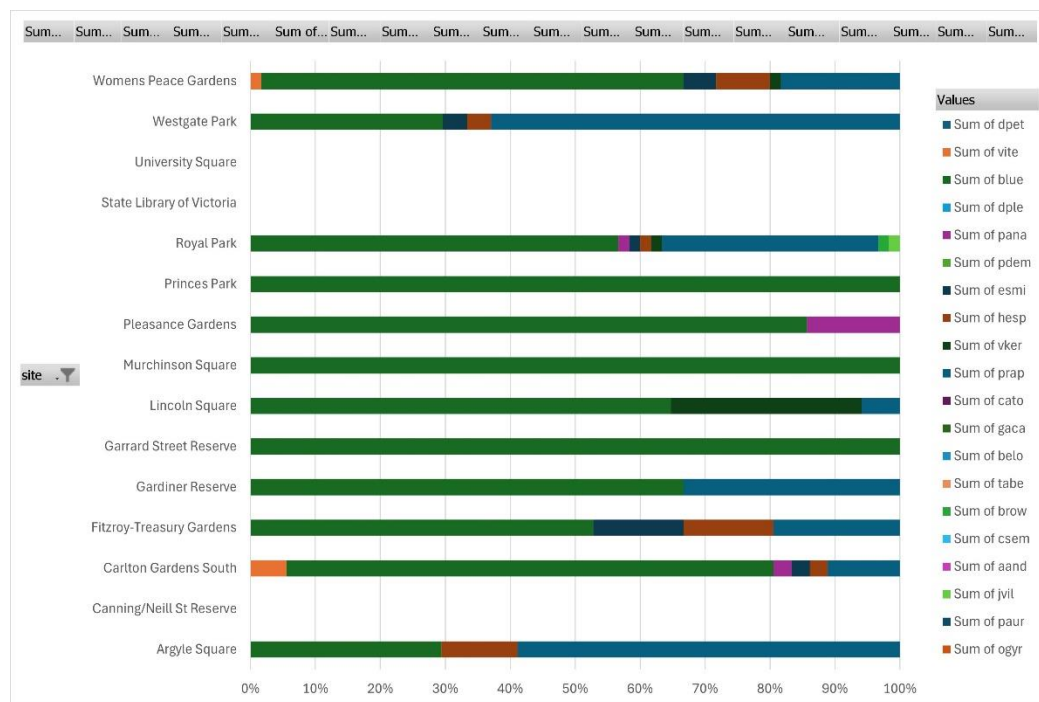


Figure 6

Inference: From this graph we can see that the greatest number of butterflies were found in Royal Park and Women's Peace Garden. A significant amount of butterfly species was also found in Fitzroy-Treasure Gardens and Carlton Gardens South as well. In University Square, State Library of Victoria and Canning site no butterfly species were found.

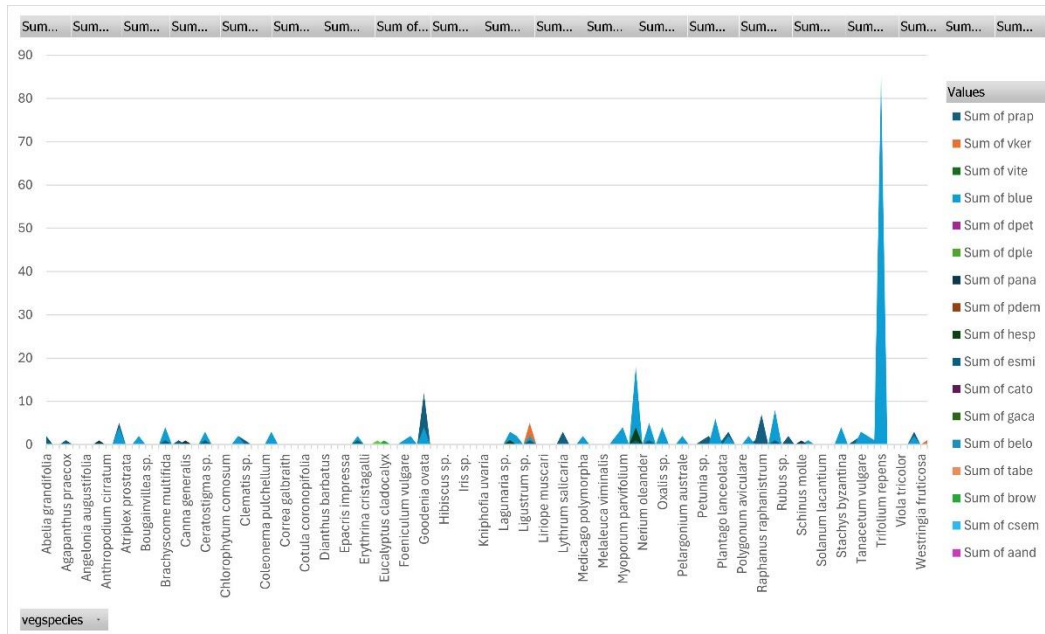


Figure 7

Inference: Across all plant species, 300 butterflies have been seen. The number of butterfly species found on various plant species varies significantly. Certain plant species draw a diverse array of butterfly species, whilst others draw either none or very few. Some kinds of butterflies appear to find plant species particularly attractive. For example, "Trifolium repens" attracted 88 butterflies of a particular species, suggesting a close relationship between that plant and that butterfly.

Conclusion

According to our data, there are notable differences in the relative humidity levels around Royal Park, with measurements being higher at sites 3 and 7 and lower at sites 1, 2, 5, and 8. These variations point to various microclimates that affect the habitat and plant diversity of butterflies. *Pieris Rapae* butterflies belong to the Asteraceae family, which is also represented in the Fabaceae family. The most sightings are recorded on Walk 1, suggesting possible hotspot locations. Variations in temperature indicate seasonal patterns, which impact microclimates. Correlations between flora variety and ecosystem size are implied by diverse walk times. Many butterfly species may be found in Royal Park and the Women's Peace Garden, underscoring the importance of understanding the interaction between plants and butterflies for ecosystem management and conservation.

References

Authors, Holly Kirk, Tessa Smith, Anna Backstrom, Alejandra Morán-Ordóñez, Georgia E. Garrard, Ascelin Gordon, Christopher D. Ives, Sarah Bekessy & Luis Mata. (2017). “Our City’s Little Gems
Butterfly diversity and flower-butterfly interactions in the City of Melbourne”.

Dataset Used: <https://data.melbourne.vic.gov.au/explore/dataset/butterfly-biodiversity-survey-2017/information/?location=11,-37.86613,144.86445&basemap=mbs-7a7333>

<https://rmit.instructure.com/courses/124253/files/37169778?wrap=1>

Literature Review:

<https://rmit.instructure.com/courses/124253/files/37169273?wrap=1>

Other information:

<https://www.ecoredux.com/butterflies-are-important>